

induction

Scientists in the 1820s realised that relative **motion of a conductor** in a **magnetic field** leads to the production of an **induced emf** and to an **induced current** in the conductor.

This introduced new concepts that helped terrorise G12 students: **magnetic flux**, **magnetic flux density**, and **magnetic flux linkage**.

Magnetic flux Φ is proportional to the number of field lines passing through a surface.

The unit of magnetic flux is the **weber** (Wb).

Magnetic flux is the **product** of the **magnetic flux density** B and area A , so $\Phi = B \times A$, where area A is the area through which **the conductor moves** and the **normal** to that area is **parallel** to the **direction of the magnetic field**.

Not that B is also the same B as the [magnetic field strength](#); magnetic flux density **is** magnetic field strength

The units of **magnetic flux density** is expressed as Wb m^{-2} as well as in tesla (T); $1\text{T} \equiv 1\text{Wb m}^{-2}$.

In the formula book of depression, $\Phi = BA \cos(\theta)$ as it accounts for when $\exists$ an angle θ between the **magnetic field** and the **normal** to the area A .

TODO: to be continued...